

Daniel Chigbu

07599 999871 | d.j.chigbu@gmail.com | linkedin.com/in/danchigbu | github.com/ch1gg5 | ch1gg5.github.io

EDUCATION

Loughborough University

Bachelor of Science with Honours in Computer Science

- *First Year Grade: First Class Honours*

Leicestershire, UK

Sept. 2025 – June 2029

PROJECTS

FocusFlow API | *Python, FastAPI, PostgreSQL, SQLAlchemy, JWT*

July 2026

- Developed a RESTful productivity analytics API using FastAPI with JWT authentication, SQLAlchemy ORM and PostgreSQL.
- Designed relational data models for tasks, categories and productivity sessions with secure user-scoped access.
- Built analytics endpoints using SQL aggregation queries to calculate productivity trends, category breakdowns and study streaks.

Coursework Deadline Optimiser | *Java, Spring Boot, Spring Security, JWT, PostgreSQL, JPA*

June 2026

- Built and deployed a full-stack coursework management application using Spring Boot, PostgreSQL and JWT authentication with a layered Controller–Service–Repository architecture.
- Developed a scheduling algorithm that prioritises study time based on deadline urgency, task difficulty and course weighting

Student Performance Predictor - ML Project | *Python, Pandas, NumPy, Scikit-learn, Matplotlib*

May 2026

- Built an end-to-end machine learning regression pipeline to predict student exam scores from 13 numerical and categorical features
- Compared and tuned four regression models - Linear Regression, Decision Tree, Random Forest and Gradient Boosting - using MAE, RMSE, R^2 and 5-fold cross-validation to evaluate generalisation.
- Selected Linear Regression as the final model, achieving $R^2 = 0.825$ and $MAE = 0.415$ on the held-out test set; performed residual and feature analysis to identify model limitations and unexplained high-score predictions.

Loughborough University Route Finder | *Python, A* Search, Matplotlib, NetworkX*

May 2026

- Implemented A* pathfinding algorithm with Euclidean heuristic for optimised route computation across campus with obstacle avoidance and multi-waypoint support
- Built graph visualisation system using NetworkX and Matplotlib; mapped nodes with coordinates and edge weights for route analysis

HACKATHONS

RoomQuest | *Google Antigravity, Gemini API, React, TypeScript*

Google Student AI Hackathon, London

- Selected as 1 of 52 participants from 1000 applicants for a Google-run AI hackathon; acted as project manager for a 5-person team, defining product scope, system architecture and tech stack under a 3-hour build constraint.
- Directed AI-assisted development through an AI Studio → Antigravity → Google Cloud workflow, coordinating task delegation to ship a working demo that generates room-aware puzzle clues via Gemini multimodal vision in under 5 seconds.
- Built and presented the accompanying business case, sizing the market opportunity (\$14.8B TAM, \$3.53B SAM) and defining a go-to-market strategy alongside the technical demo.

EXPERIENCE

JP Morgan Chase Software Engineering

July 2026

Job Simulation

Forage

- Integrated Kafka into a Spring Boot microservice to consume and deserialize high-volume transaction messages using a configurable topic and embedded Kafka test framework.
- Developed a REST endpoint for querying user balances, returning JSON responses through a Spring controller while maintaining clean architectural boundaries.

Quantum Software Engineering

June 2026

Job Simulation

Forage

- Developed an interactive Dash application that enabled the client to assess the impact of price changes on sales and profitability.
- Implemented a test suite to verify the Dash application is working and a bash script to automatically run the test suite.

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, HTML, CSS

Frameworks & APIs: Spring Boot, Spring Security, JPA/Hibernate, FastAPI, React, Apache Kafka, Gemini API

Databases & Cloud: PostgreSQL, MySQL, SQLite, Render, Vercel

Developer Tools: Git, GitHub, GitHub Actions, Docker, JUnit, PyTest, Postman, IntelliJ IDEA, PyCharm, Maven

Libraries: JDBC, NetworkX, Matplotlib, Pandas, Dash, JWT, SQLAlchemy